SHAIKH JINNATUL

B.Sc. Microbiology (Honours)

Microbiology | Molecular Biology | Laboratory Research

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PROFESSIONAL SUMMARY

Motivated Microbiology (Honours) student (Major CGPA 9.07/10, graduating 2027) with hands-on laboratory experience in water microbiology, microbial isolation and culture, antimicrobial susceptibility testing, bacterial transformation and SDS-PAGE. Skilled in aseptic technique, careful observation and accurate data recording. Seeking an entry-level role in microbiology, molecular biology or laboratory research, with interests in microbial biotechnology, plant–microbe interactions and antimicrobial resistance.

EDUCATION

B.Sc. Microbiology (Honours), 4-Year, NEP 2020 — **Expected 2027**

MUC Women's College, The University of Burdwan · **Major CGPA 9.07 · All-subject CGPA 8.90**

ISC (Class XII) · East West Model School — **91.80%**

ICSE (Class X) · East West Model School — **91.08%**

PROJECTS

Bacterial Transformation and Transformation Efficiency

- Performed bacterial transformation with plasmid DNA; plated cells on LB agar containing X-gal, IPTG and antibiotic.
- Applied blue-white screening to distinguish colony phenotypes and calculated transformation efficiency.

Microbiological Analysis of Water Using MPN and Membrane Filtration

- Carried out presumptive, confirmed and completed MPN tests on water samples from a pond and an Aquaguard water purifier.
- Applied membrane filtration for microbiological analysis of water.

Antibiotic Sensitivity Testing of *E. coli* and *Bacillus* Species

- Performed agar cup (well diffusion) assays using streptomycin, ampicillin, azithromycin and kanamycin, with a control.
- Measured and marked zones of inhibition to compare antibiotic susceptibility between the two organisms.

Isolation of *Rhizobium* from Soil and Carrier-Based Biofertiliser Production

- Worked on the isolation of *Rhizobium* from soil samples.
- Participated in carrier-based *Rhizobium* biofertiliser production in the college laboratory (Semester VII).

SDS-PAGE Analysis

- Separated protein samples by molecular weight alongside a protein marker.
- Stained the gel and analysed banding patterns to compare samples.

TECHNICAL SKILLS

Culture and Isolation	Culture on nutrient, blood, chocolate, EMB and MacConkey agar and in nutrient and lactose broth; slant and stab culture preservation; isolation of *Rhizobium* and spoilage microorganisms.
Staining and Microscopy	Gram, simple, LPCB and acid-fast staining; light microscopy.
Water and Antimicrobial Testing	MPN test (presumptive, confirmed, completed); membrane filtration; agar cup (well diffusion) assay.
Molecular Biology	Bacterial transformation, blue-white screening, transformation efficiency calculation; SDS-PAGE; genomic DNA isolation.
Food, Dairy and Biochemistry	Probiotic foods (kanji, dahi); MBRT of milk; alkaline phosphatase test; estimation of carbohydrates, protein and vitamin C.
Aseptic Practice and Biosafety	Aseptic technique; autoclave and hot-air oven sterilisation; filtration of heat-sensitive materials; laminar airflow cabinet and incubator; laboratory biosafety.
Observed (Exposure)	Disc diffusion and Kirby–Bauer assays; MIC of streptomycin against *E. coli*; plasmid DNA isolation and agarose gel electrophoresis; DOT-ELISA; leukocyte counts; TMV infection study; soil analysis.
Microbial Biotechnology	Carrier-based *Rhizobium* biofertiliser production; principles of mushroom spawn production; probiotic food analysis.
Data and In-Silico Skills	Measurement of inhibition zones; transformation efficiency calculation; analysis of SDS-PAGE banding patterns; primer design and in-silico validation; MS Excel.
Computer Skills	MS Word, MS Excel, MS PowerPoint

CERTIFICATIONS AND TRAINING

One-Day Intensive Workshop on PCR and Primer Designing with In-Silico Validation
Flavin Labs Pvt. Ltd. | July 2025
Covered PCR principles and primer design, with in-silico validation of primers before use.

Hands-on Training on "Basics of Molecular Biology"
Oriental Institute of Science and Technology (Vidyasagar University) | May 2025
Practical two-day training in the core techniques and concepts of molecular biology.

3-Day Online Workshop on Mushroom Spawn Production
Flavin Labs Pvt. Ltd. | September 2025
Learned the principles and stages of mushroom spawn production in a three-day online workshop.

LANGUAGES

English · Bengali · Hindi · Urdu